ALPhA-Vision: A Real-Time Always-On Vision Processor with 787µs Face Detection Latency in <5mW

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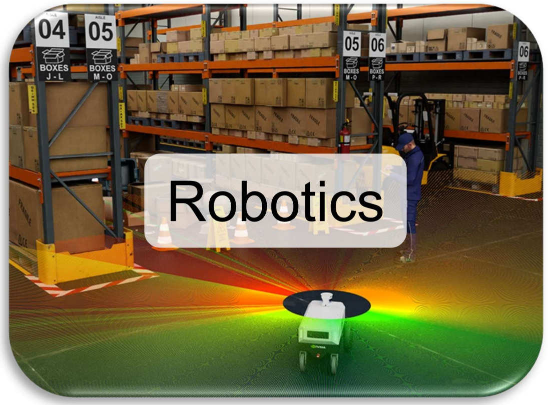
Outline

- **Motivation and Design Principles**
- Design Overview and Key Features
- Testchip Measurement Results

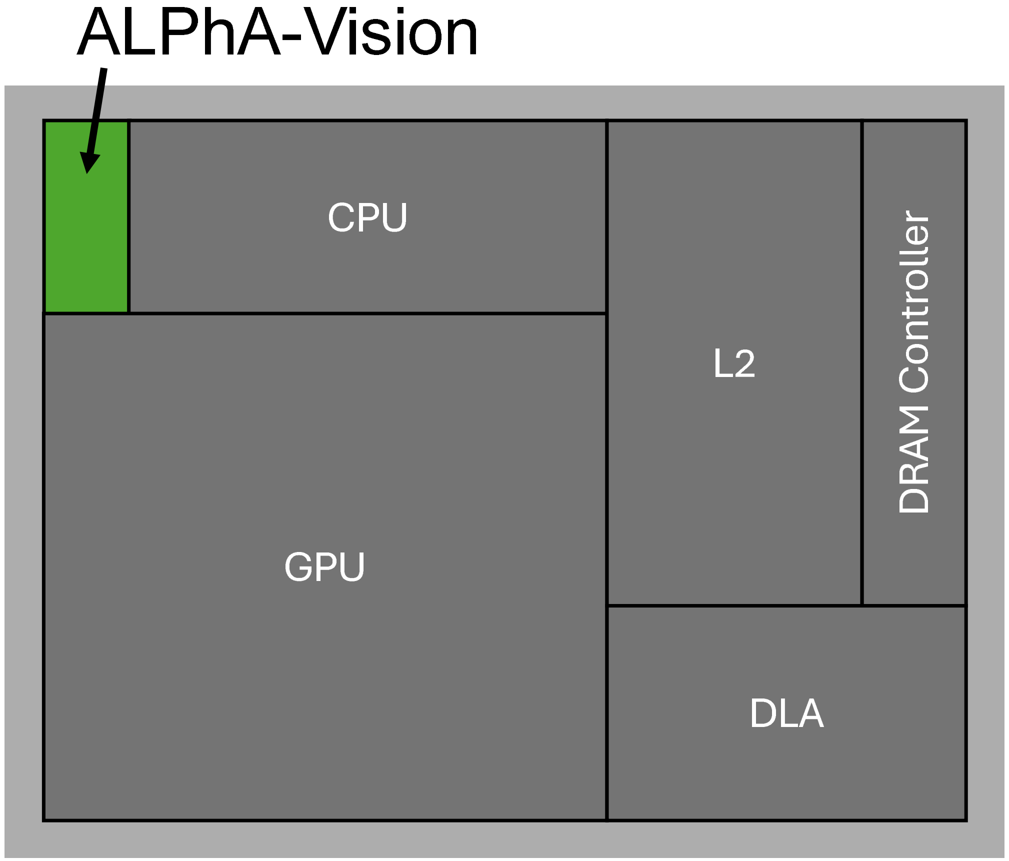
Motivation: An Always-on Future



- Application requirements:
  - Always-on vision
  - Low latency
  - Low power

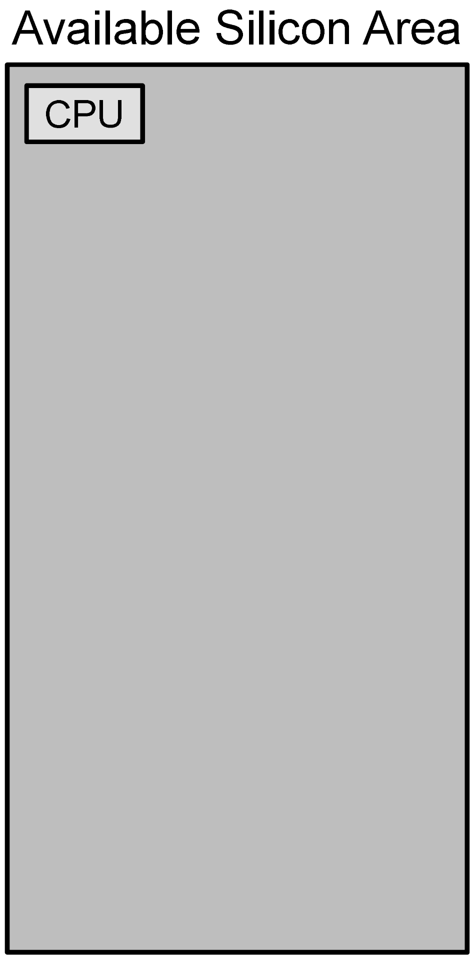
Solution: Always-on Vision Subsystem

- **Tiny subsystem** integrated on main SoC
- **Tiny (<10mW) always-on power overhead** during continuous operation
- **Large system-level power savings** from opportunistic sleep of displays/actuators and other SoC components



Challenge 1: Large Memory Footprint

- All weights and activations must fit in local memory



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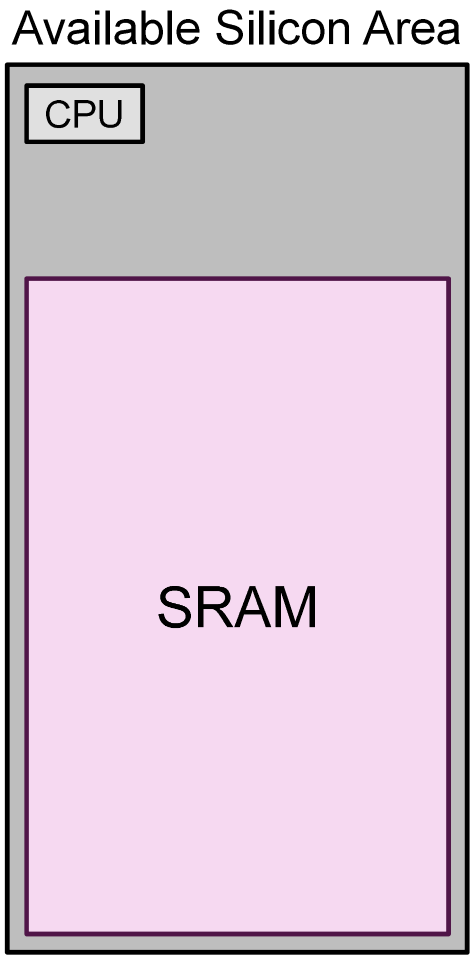
- Solution 1: Fill most of the silicon area with SRAM backing storage (~2MB)

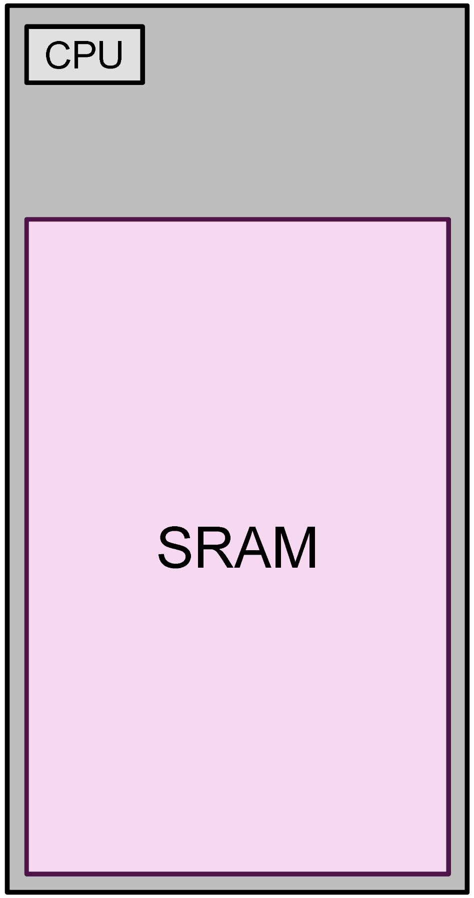


Challenge 1: Large Memory Footprint

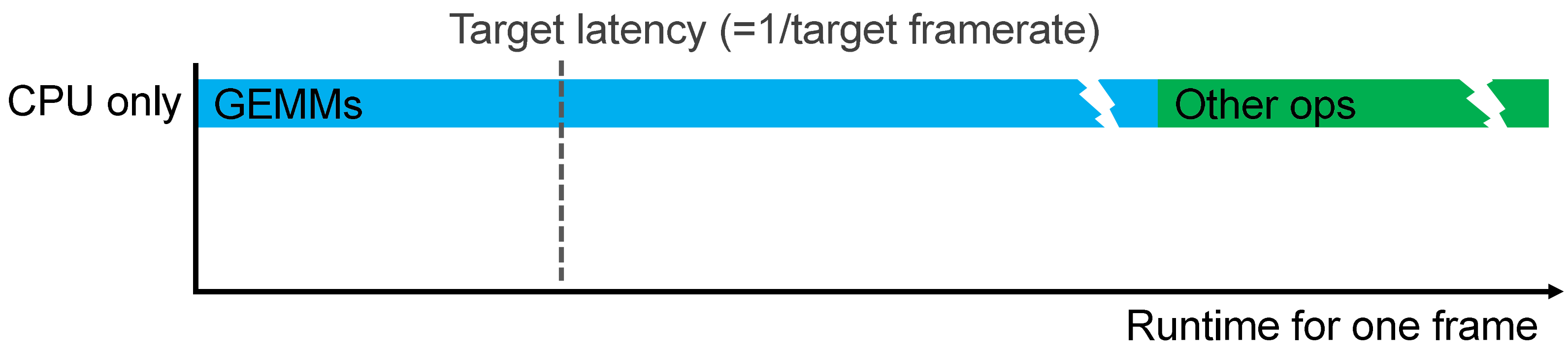
- All weights and activations must fit in local memory
- Solution 1: Fill most of the silicon area with SRAM backing storage (~2MB)
- Solution 2: Aggressive quantization with hardware-aware QAT to recover accuracy

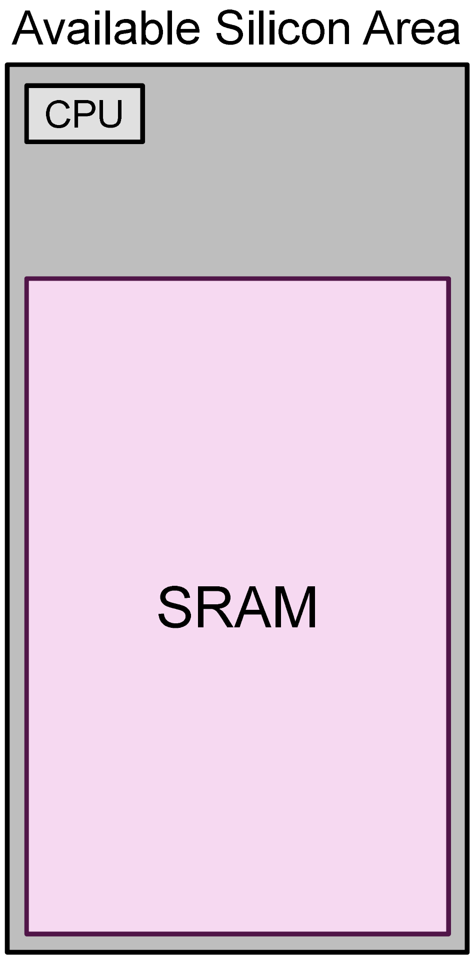
Yolov5n-0.5 Face Detection		
	FP32	**Hybrid INT8/INT4**
Weight Memory (MB)	1.7	**0.38**
Peak Activation Memory (MB)	2.3	**0.46**
FDDB Detection Rate	100	**99.3**

Available Silicon Area



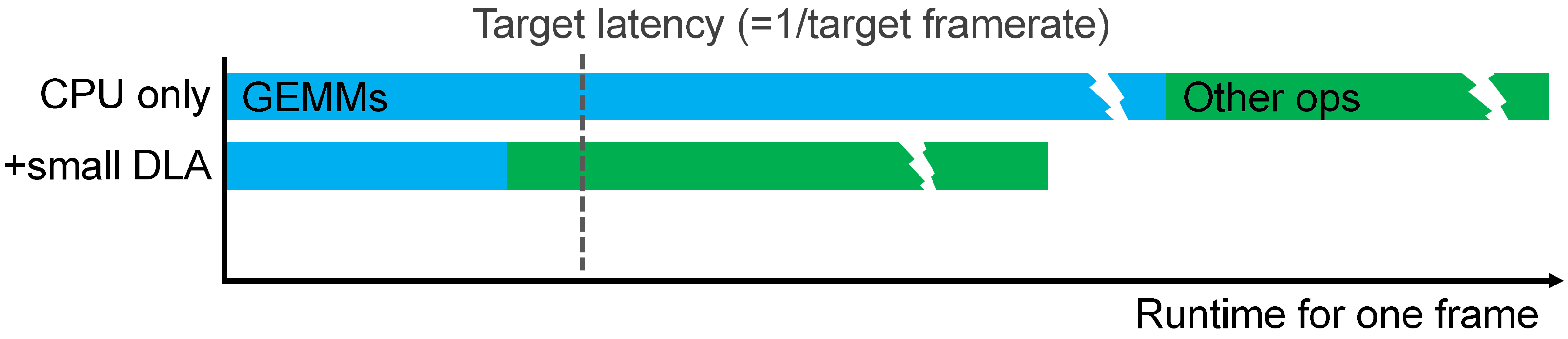
Challenge 2: Diverse Workloads

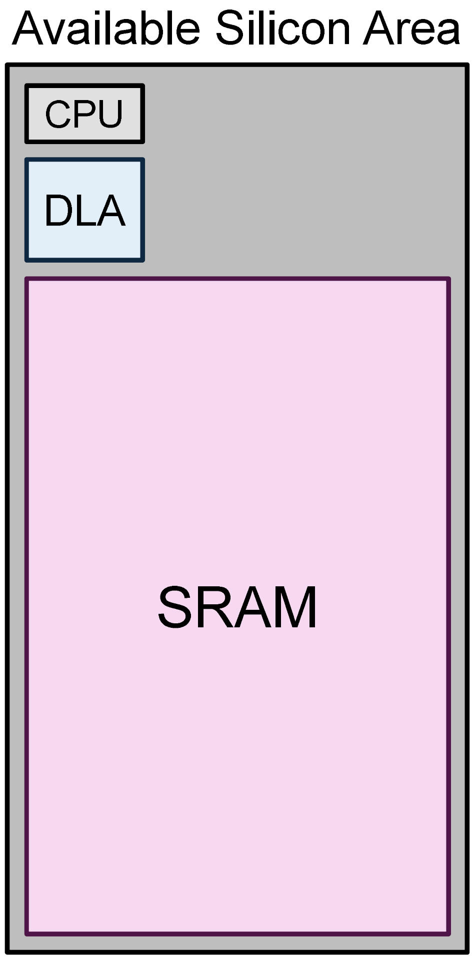




- Need hardware specialization to meet tight latency constraint (60fps -> 16.7ms)

Challenge 2: Diverse Workloads





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- Solution 1: Add deep learning accelerator (DLA) to accelerate GEMMs (most ops)

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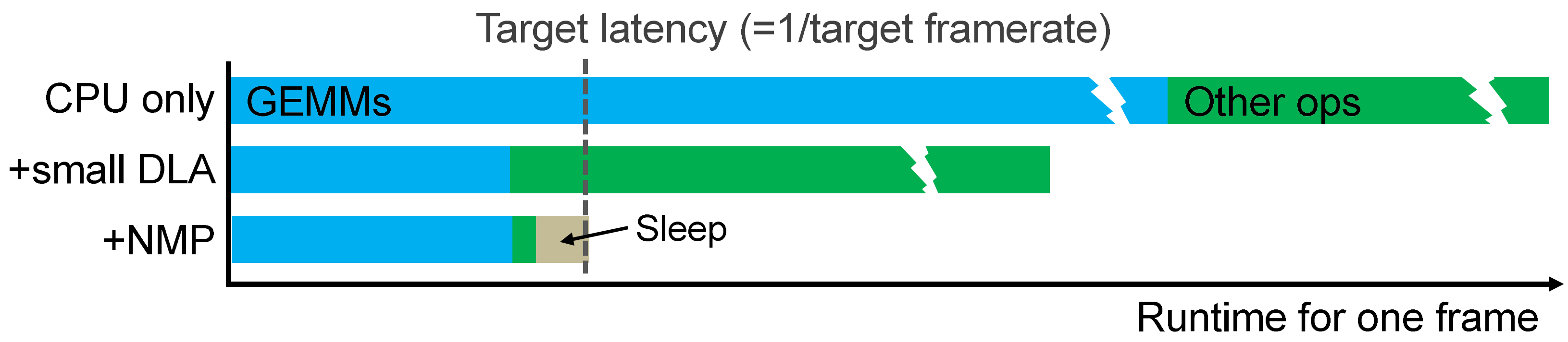
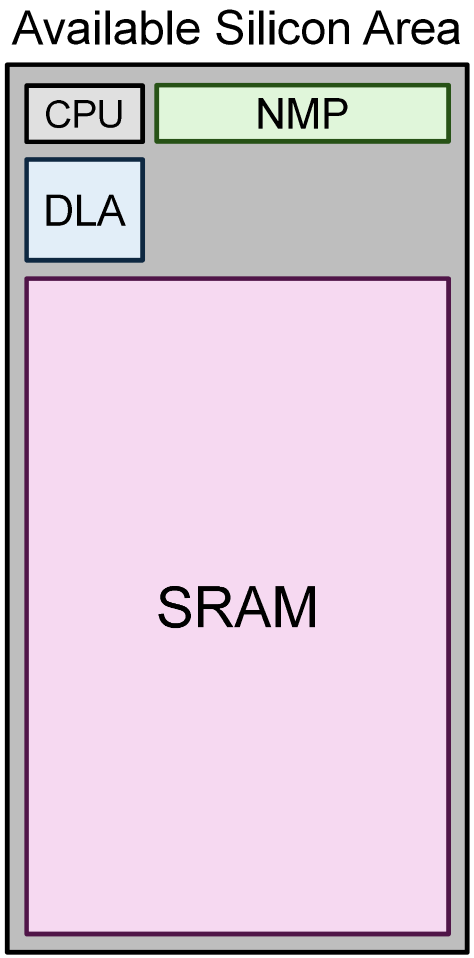
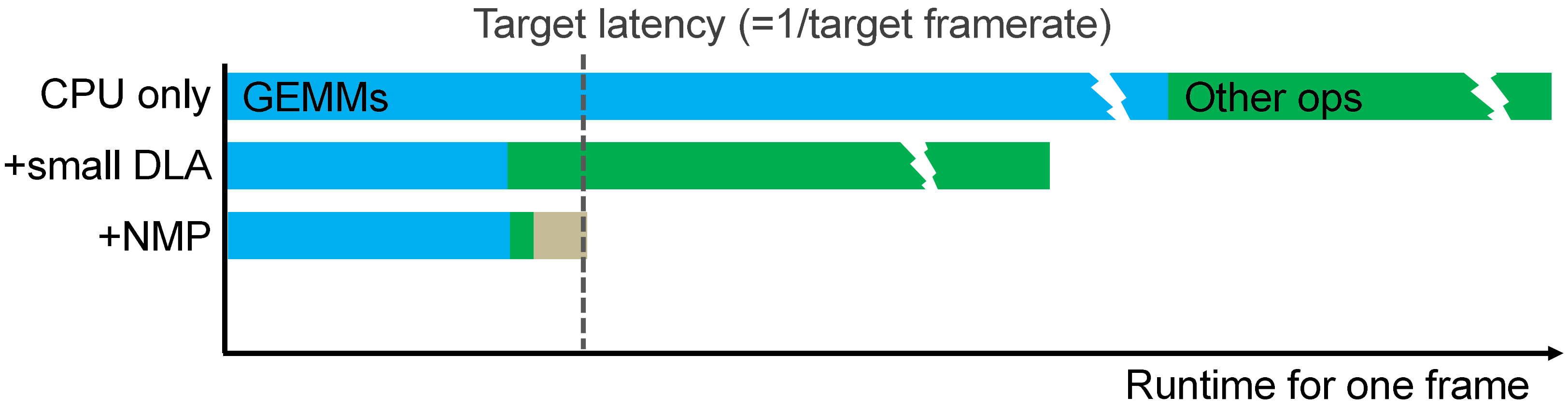
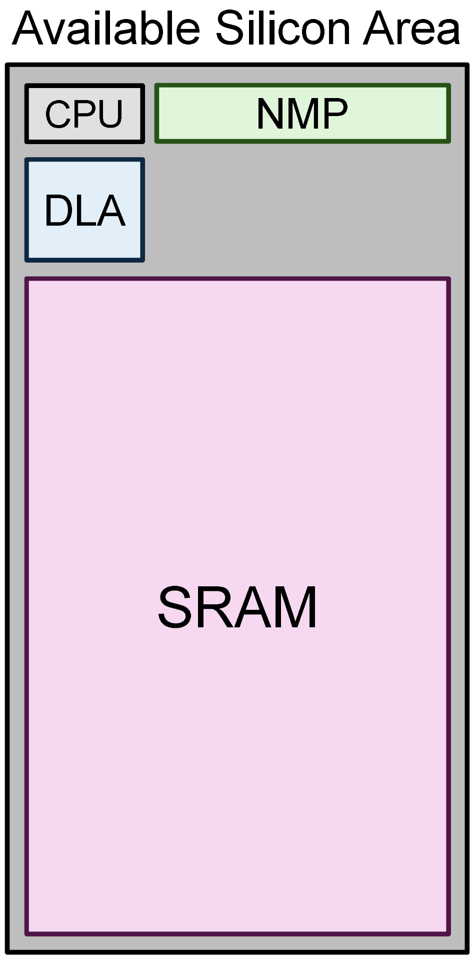




- Need hardware specialization to meet tight latency constraint (60fps -> 16.7ms)
- Solution 1: Add deep learning accelerator (DLA) to accelerate GEMMs (most ops)
- Solution 2: Add near-memory processor (NMP) to accelerate "long tail" of non-GEMM ops

Challenge 3: Leakage Power





- SRAM leakage dominates always-on power

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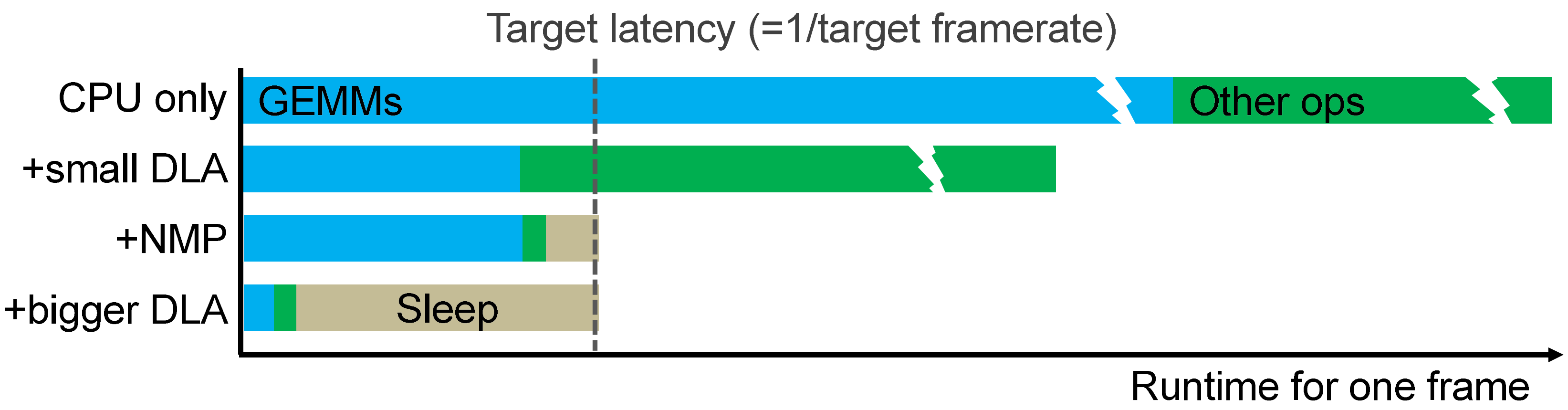
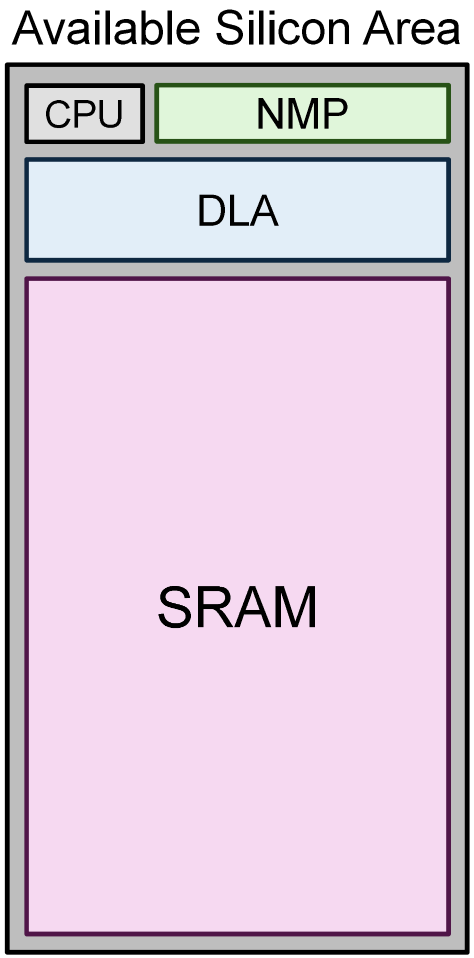
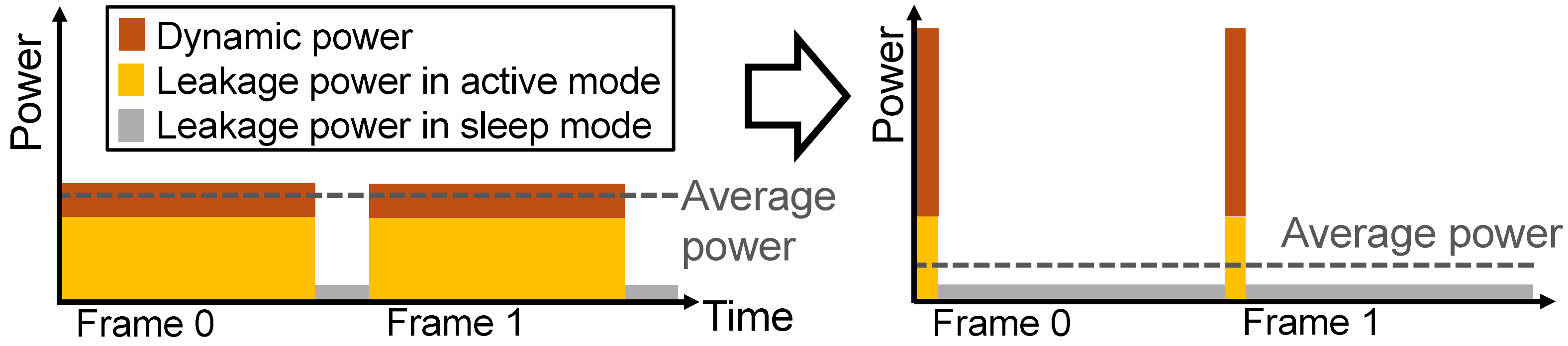




- SRAM leakage dominates always-on power
- Solution: Overprovision DLA and race to sleep



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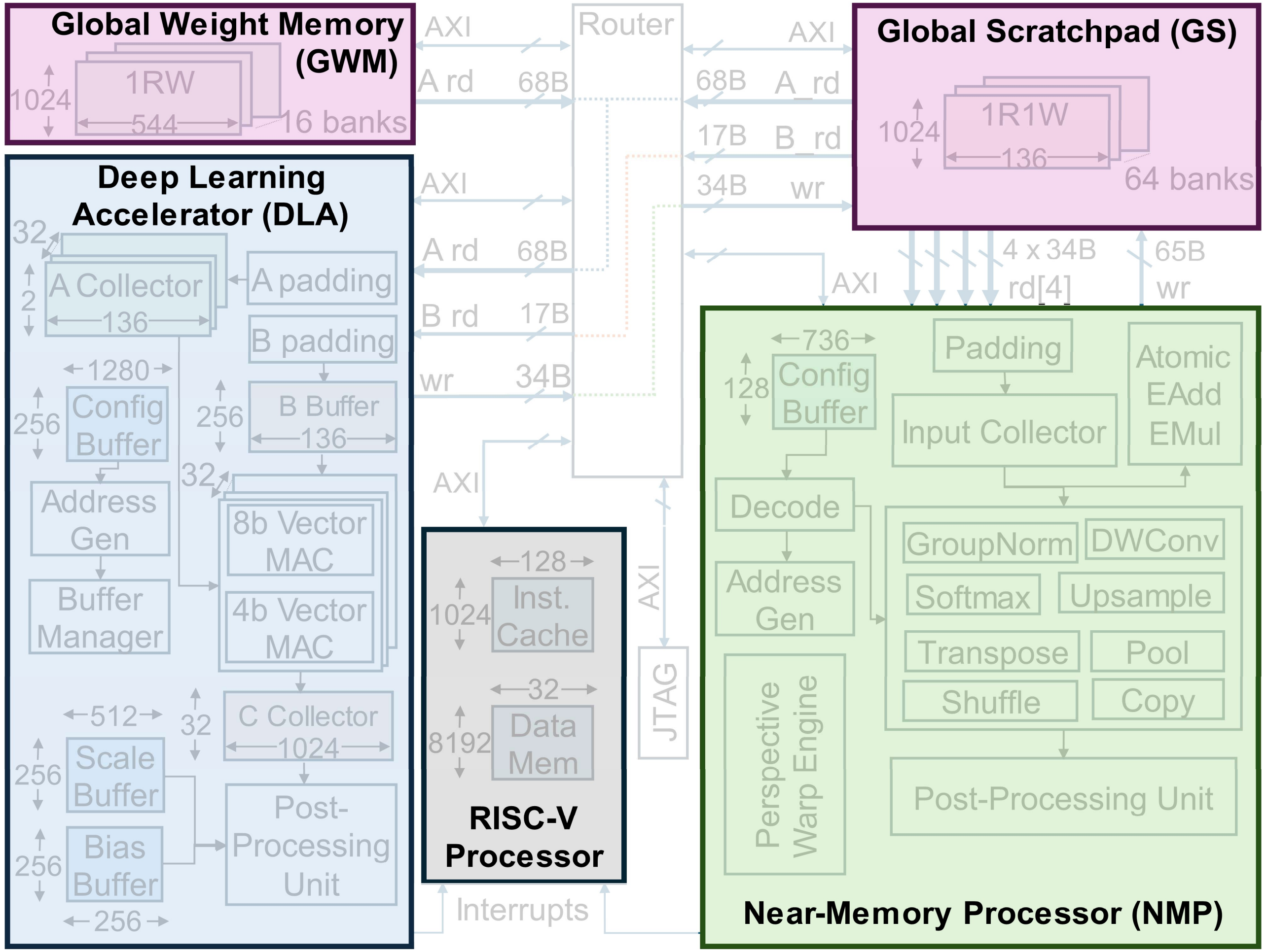
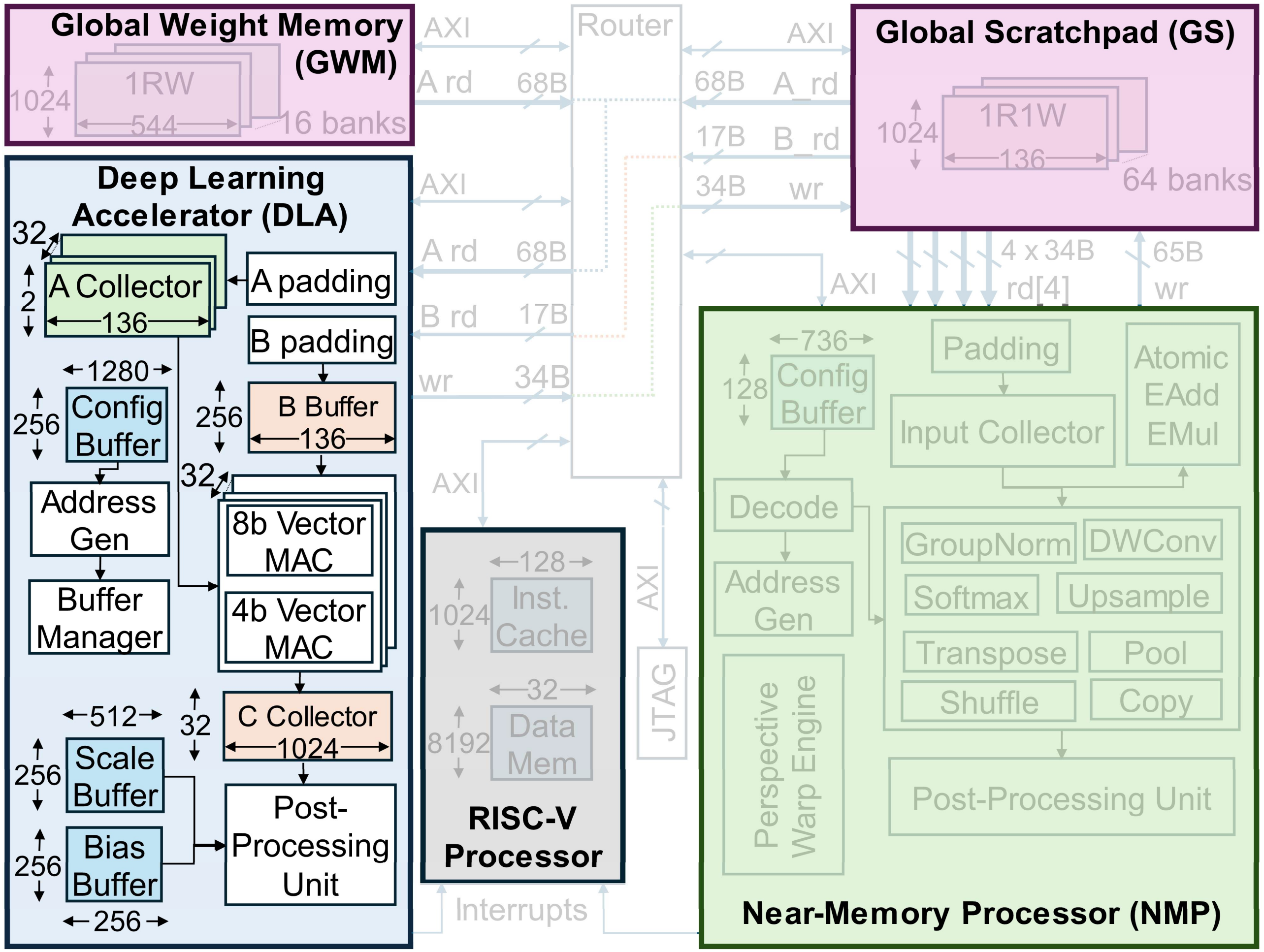
ALPhA-Vision Block Diagram



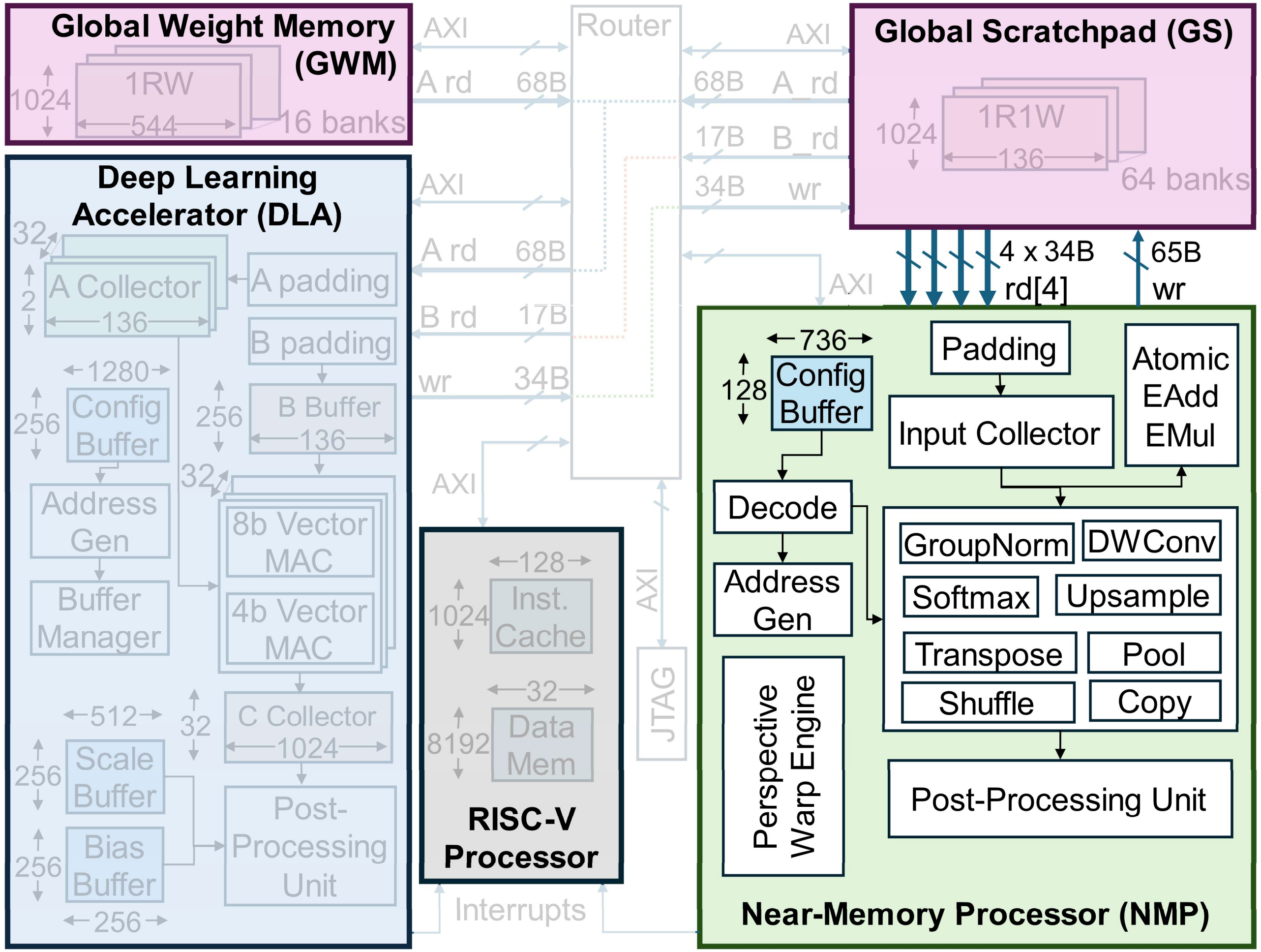
Deep Learning Accelerator

- Two data formats:
  - 512 MACs/cycle INT8
  - 1024 MACs/cycle INT4 with microscaling
- Configurable for a wide range of GEMM shapes and access patterns
- Post-processing unit performs simple non-linear operations and data format conversions



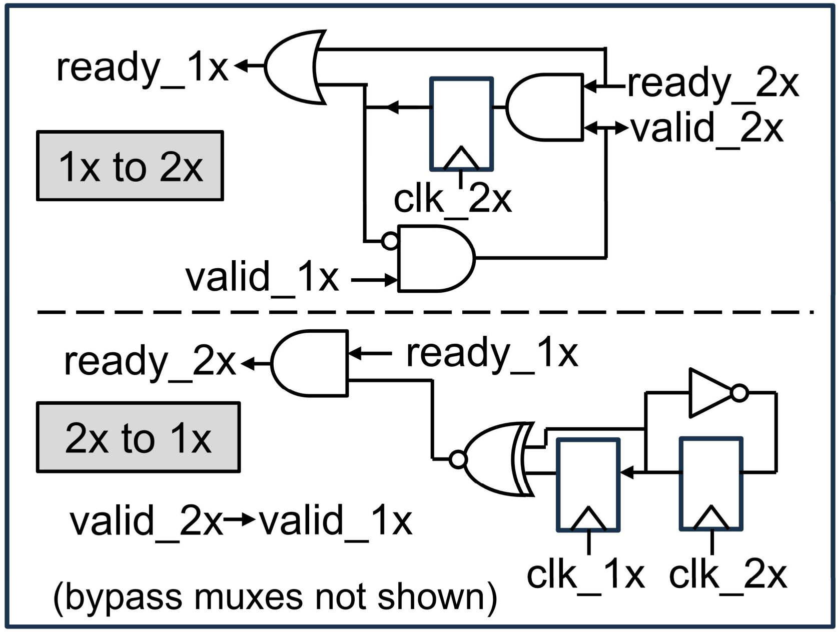
Near Memory Processor

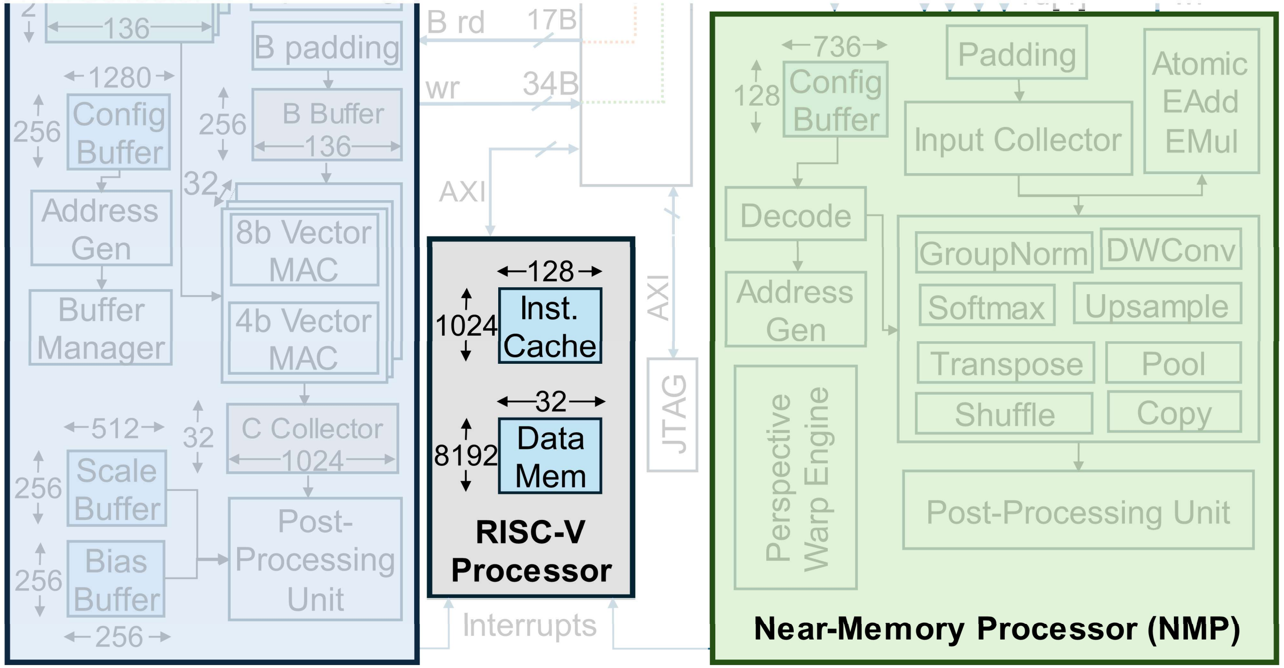
- Accelerates operations that map poorly to DLA:
  - Depthwise convolution
  - GroupNorm
  - Softmax
  - Elementwise ops
  - Memory manipulations
- High bandwidth to global scratchpad
- Integer or fixed-point data formats



Double-Clocked RISC-V Processor

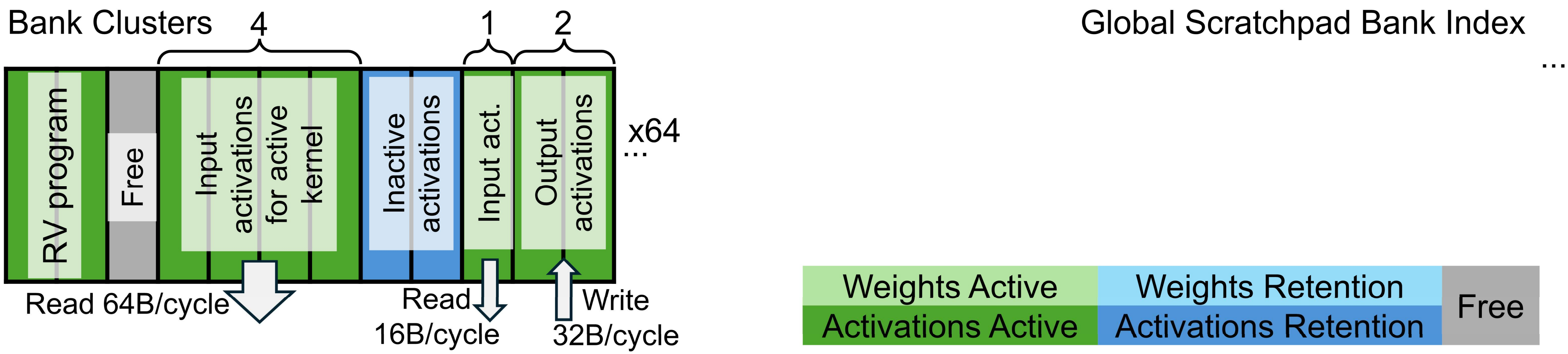
- RV clocked at 2X frequency reduces non-accelerated latency penalty
- Synchronous ready-valid interface, zero-cycle clock crossing latency
- 40% reduction in non-accelerated wall clock time





Global Scratchpad with Fine-Grained Sleep States

Active Mode	Retention Mode	Sleep Mode
	Periphery Gated, ~3x leakage reduction	Fully Gated, ~100x leakage reduction



- Sleep mode control of 64 banks at kernel granularity
- Bank clustering trades off bandwidth for leakage
- Kernel fusion trades off control overhead for fine-grained sleep

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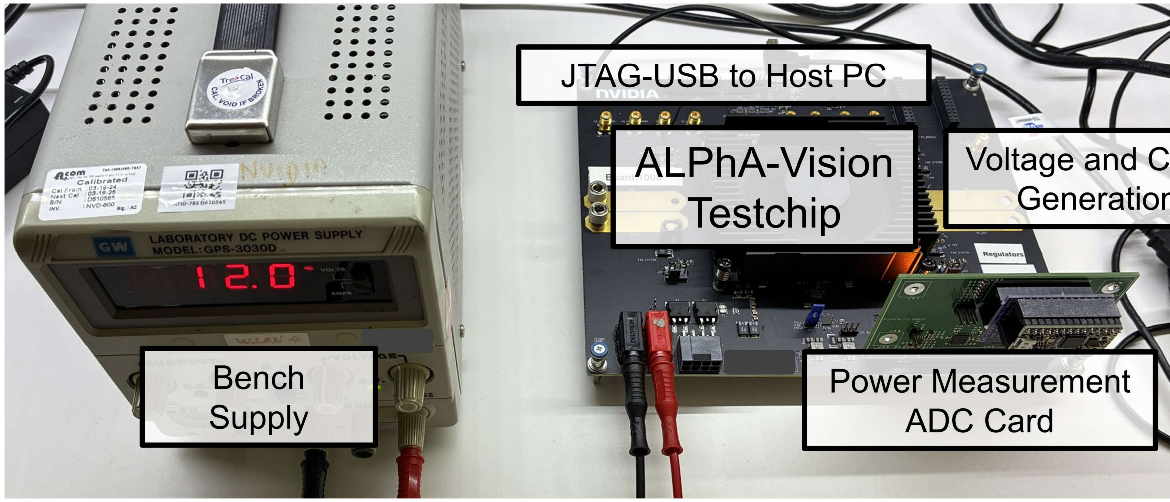
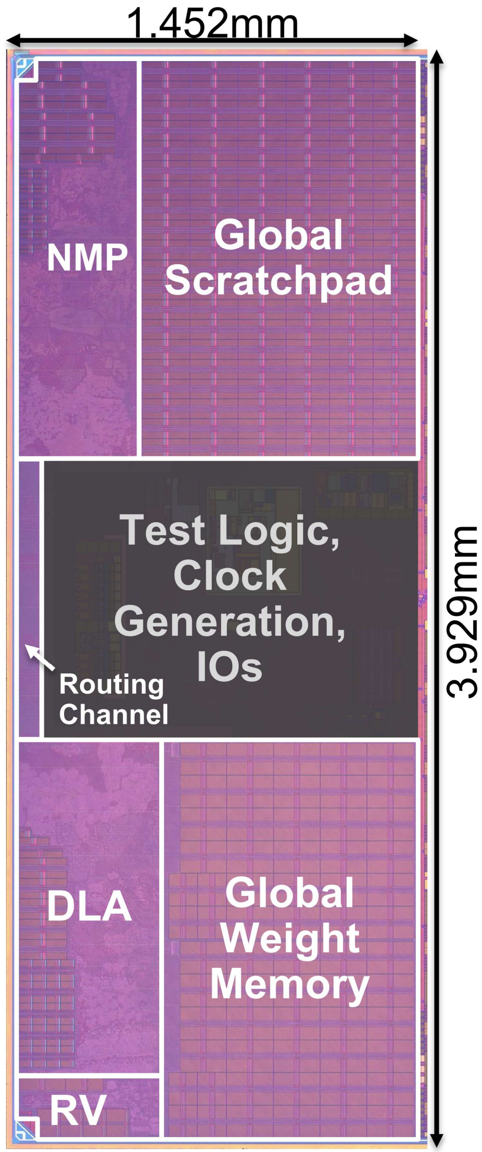
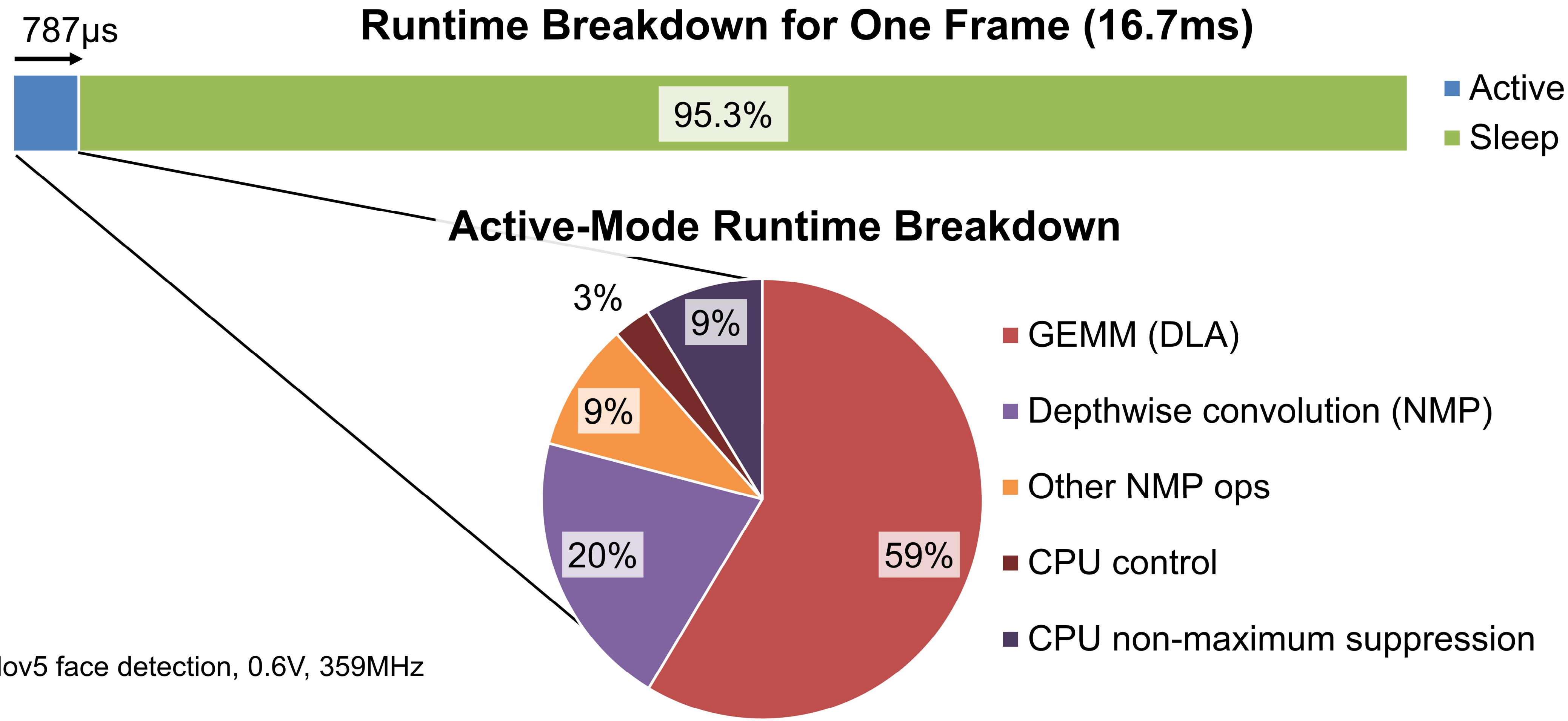
Testchip Implementation

- 4.2mm^2 prototype in 16nm
- Moderate frequency target (359MHz @ 0.6V)
  - Support race-to-sleep while avoiding gate upsizing
- >80% HVT to minimize standard cell leakage



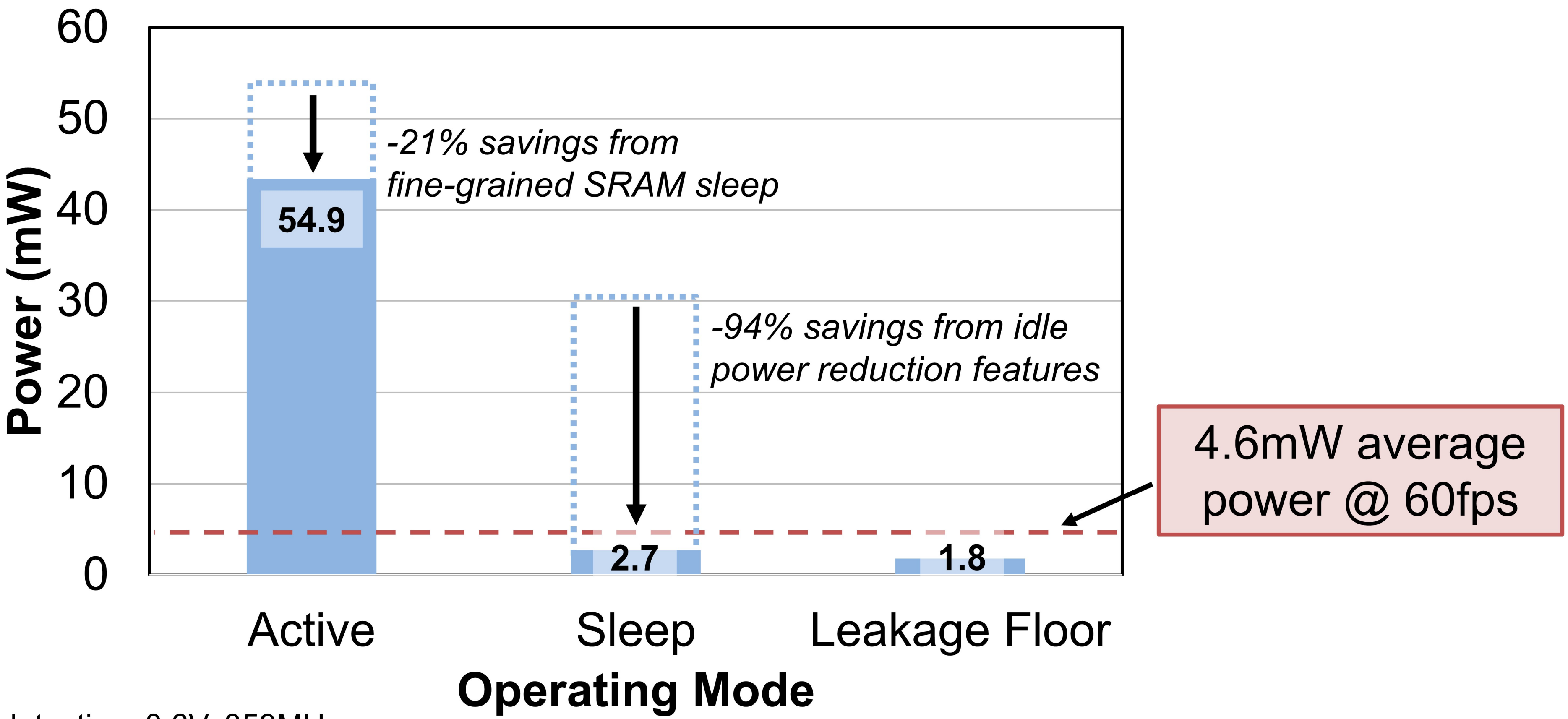


Measurement Results: Latency



Yolov5 face detection, 0.6V, 359MHz

Measurement Results: Power



Yolov5 face detection, 0.6V, 359MHz

Comparison to Prior Work

	This Work	[1]	[2]	[3]	[4]
Application	Always-on vision	Always-on vision	Always-on vision	Low-power inference	Face analysis
Process (nm)	16	40	12	28	22
Area (mm^2)	4.20	30	1.76	19.95	3.4
V_{DD} (V)	0.52-0.9	0.6-1.1	0.6	1	0.59-0.72
Frequency (MHz)	90-830	350	100-200	62.5	180-220
Power (mW)	3.9-11.5[1]	2.1[4]	0.62-1.61[5]	19-32[6]	0.16-12.77[7]
Throughput (GOPS)	1465[2]	2250	51	129	5
Latency (ms)	**0.787**[1,3]	33[4]	67-500[5]	-	50-1000[7]
Energy/frame (µJ)	**76.7**[1,3]	91[4]	107-310[5]	-	409[7]
FDDB Face Detection Rate	**99.3%**	-	-	91.3%	98%
End-to-End Execution	Yes	Yes	No	No	Yes

One MAC = 2 ops, other ops ignored. [1]Yolov5 face detection. [2]0.9V, 830MHz. [3]0.6V, 359MHz. [4]SqueezeNet image classification. [5]MobileNet human detection. [6]Workload unspecified. [7]BDT face detection + CNN face recognition.

Conclusion and Summary

- **ALPhA-Vision** is a prototype low-power subsystem for always-on vision tasks
- Key design principles:
  - **Aggressively quantize** to store all data locally
  - **Accelerate every kernel** (not just GEMMs)
  - Overprovision math and **race to sleep** to save leakage power
- The testchip prototype performs face detection with **99.3% accuracy** using **4.6mW average power** at **60fps**

References

[1] A. Gupta et al., "CogniVision: End-to-End SoC for Always-on Smart Vision with mW Power in 40 nm," IEEE Symp. VLSI Circuits, 2024. https://doi.org/10.1109/VLSITechnologyandCir46783.2024.10631426

[2] E. Chang et al., "A 12-nm 0.62-1.61 mW Ultra-Low Power Digital CIM- based Deep-Learning System for End-to-End Always-on Vision," IEEE Symp. VLSI Circuits, 2023. https://doi.org/10.23919/VLSITechnologyandCir57934.2023.10185296

[3] S. Kwon et al., "Monolithic in-Memory Computing Microprocessor for End-to-End DNN Inferencing in MRAM-Embedded 28 nm CMOS Technology with 1.1 Mb Weight Storage," ISSCC, pp. 610–612, 2025. https://doi.org/10.1109/ISSCC49661.2025.10904609

[4] P. Jokic et al., "A Sub-mW Dual-Engine ML Inference System-on-Chip for Complete End-to-End Face-Analysis at the Edge," IEEE Symp. VLSI Circuits, 2021. https://doi.org/10.23919/VLSICircuits52068.2021.9492401

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